



Helping the Animals That Need It Most

Austin Animal Center



Kayla Ryan

The Waiting Game

Austin Animal Center operates as a No-Kill shelter (95%+ live release). Kennels stay full. Resources are limited.

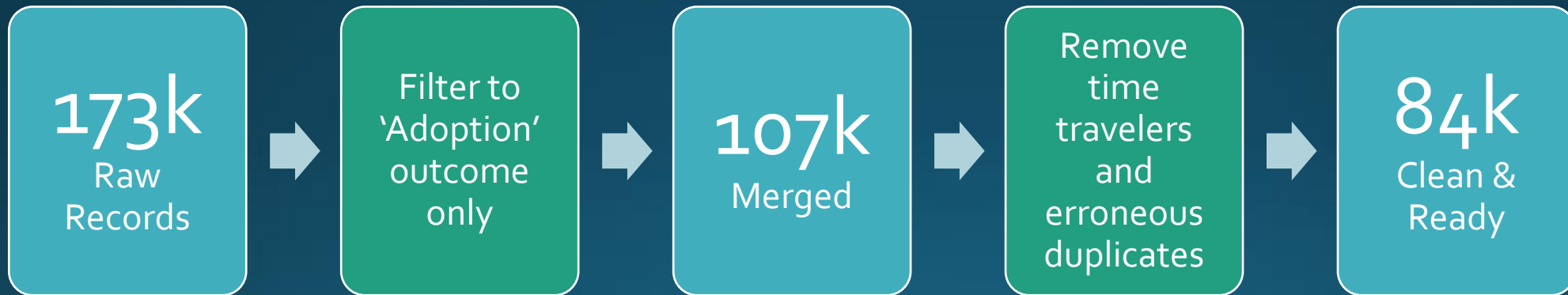
Currently, shelters operate reactively, waiting 60 days or more to realize an animal is struggling to get adopted

Day 1

Identify animals with a “Critical” risk of being a long-term resident the moment they come in the door

Cleaning the Noise

Raw shelter data is messy. I spent a lot of time cleaning and preprocessing the features to make sure it was ready for model fitting.



Data columns (total 12 columns):			
#	Column	Non-Null Count	Dtype
0	Animal ID	173812 non-null	object
1	Name	123821 non-null	object
2	DateTime	173812 non-null	object
3	MonthYear	173812 non-null	object
4	Found Location	173812 non-null	object
5	Intake Type	173812 non-null	object
6	Intake Condition	173812 non-null	object
7	Animal Type	173812 non-null	object
8	Sex upon Intake	173811 non-null	object
9	Age upon Intake	173812 non-null	object
10	Breed	173812 non-null	object
11	Color	173812 non-null	object

	Animal ID	Name	DateTime	MonthYear	Found Location	Intake Type	Intake Condition	Animal Type	Sex upon Intake	Age upon Intake	Breed	Color
0	A521520	Nina	10/01/2013 07:51:00 AM	October 2013	Norht Ec in Austin (TX)	Stray	Normal	Dog	Spayed Female	7 years	Border Terrier/ Border Collie	White/Tan
1	A664235	NaN	10/01/2013 08:33:00 AM	October 2013	Abia in Austin (TX)	Stray	Normal	Cat	Unknown	1 week	Domestic Shorthair Mix	Orange/ White
2	A664236	NaN	10/01/2013 08:33:00 AM	October 2013	Abia in Austin (TX)	Stray	Normal	Cat	Unknown	1 week	Domestic Shorthair Mix	Orange/ White
3	A664237	NaN	10/01/2013 08:33:00 AM	October 2013	Abia in Austin (TX)	Stray	Normal	Cat	Unknown	1 week	Domestic Shorthair Mix	Orange/ White
4	A664233	Stevie	10/01/2013 08:53:00 AM	October 2013	7405 Springtime in Austin (TX)	Stray	Injured	Dog	Intact Female	3 years	Pit Bull Mix	Blue/White

Raw “intake” dataset

Raw “intake” dataset datatypes

Data columns (total 82 columns):			
#	Column	Non-Null Count	Dtype
0	Animal ID	163932 non-null	object
1	i_datetime	163932 non-null	datetime64[ns]
2	i_month	163932 non-null	int32
3	i_year	163932 non-null	int32
4	i_age_days	163932 non-null	int64
5	is_cat	163932 non-null	int64
6	sex_female	163932 non-null	int64
7	sex_unknown	163932 non-null	int64
8	status_fixed	163932 non-null	int64
9	no_name	163932 non-null	int64
10	cond_injured	163932 non-null	int64
11	cond_neonatal	163932 non-null	int64
12	cond_nursing	163932 non-null	int64
13	cond_other	163932 non-null	int64
14	cond_sick	163932 non-null	int64

Cleaned/encoded dataset -82 columns!

Merge Tables

```
[39]: # merge the tables with inner join on Animal ID, creating new 'master' df
# (dropping any records from intake df whose 'Animal ID' doesn't match a record in outcome_final df (previously narrowed to adoption only outcomes) )
master = pd.merge(intake, outcome_final, on='Animal ID', how='inner')
print(master.shape)
```

(107588, 83)

Calculate Length of Stay

```
[40]: # calculate the Length of Stay ( (outcome date/time) - (income date/time) )
master['length_of_stay'] = (master['o_datetime'] - master['i_datetime']).dt.total_seconds() / 86400 # (total seconds of stay/seconds per day)
```

After tables were merged, length of stay was calculated to determine “risk” categories

long_stay_risk	
Elevated	33.957903
Nominal	30.547727
Severe	25.523259
Critical	9.971110

Stay Risk

Nominal = up to 7 days

Elevated = up to 30 days

Severe = up to 3 months

Critical = more than 3 months

Weighted for Success

- Animals staying more than 3 months make up only 10% of the records in this dataset. A standard model largely ignores them to boost its overall accuracy.
- I've weighted the categories to push for better recall in this Critical category. In this project, a false negative (overlooking a cat in need) is awful. A false positive (extra marketing for an easier to adopt dog) is not harmful.



```
# train model
clf = DecisionTreeClassifier(random_state=fold, class_weight = custom_weights)
# hyperparameter tuning parameters
grid_values = {'criterion': ['gini', 'entropy'], 'max_depth':[None, 5, 10, 25, 50]}
```

Hyperparameter tuning

```
# stratified cross validation with 5 folds
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=321)

fold = 0
f1 = []
precision = []
recall = []
accuracy = []
features = []
```

Cross-validation

```
# feature selection
f_val, p_val = chi2(train_x, train_y)

# chi2 and p values
df_scores = pd.DataFrame(zip(train_x.columns, f_val, p_val), columns=["feature", "chi2", "p"])
df_scores["chi2"] = df_scores["chi2"].round(2)
df_scores["p"] = df_scores["p"].round(3)

# use significant features (p < 0.05)
sel_features = df_scores[df_scores["p"]<0.05]["feature"].values
```

Feature selection

```
#scaling
scaler = StandardScaler()
train_x_scaled = pd.DataFrame(scaler.fit_transform(train_x[sel_features]), columns=sel_features)
val_x_scaled = pd.DataFrame(scaler.transform(val_x[sel_features]), columns=sel_features)
```

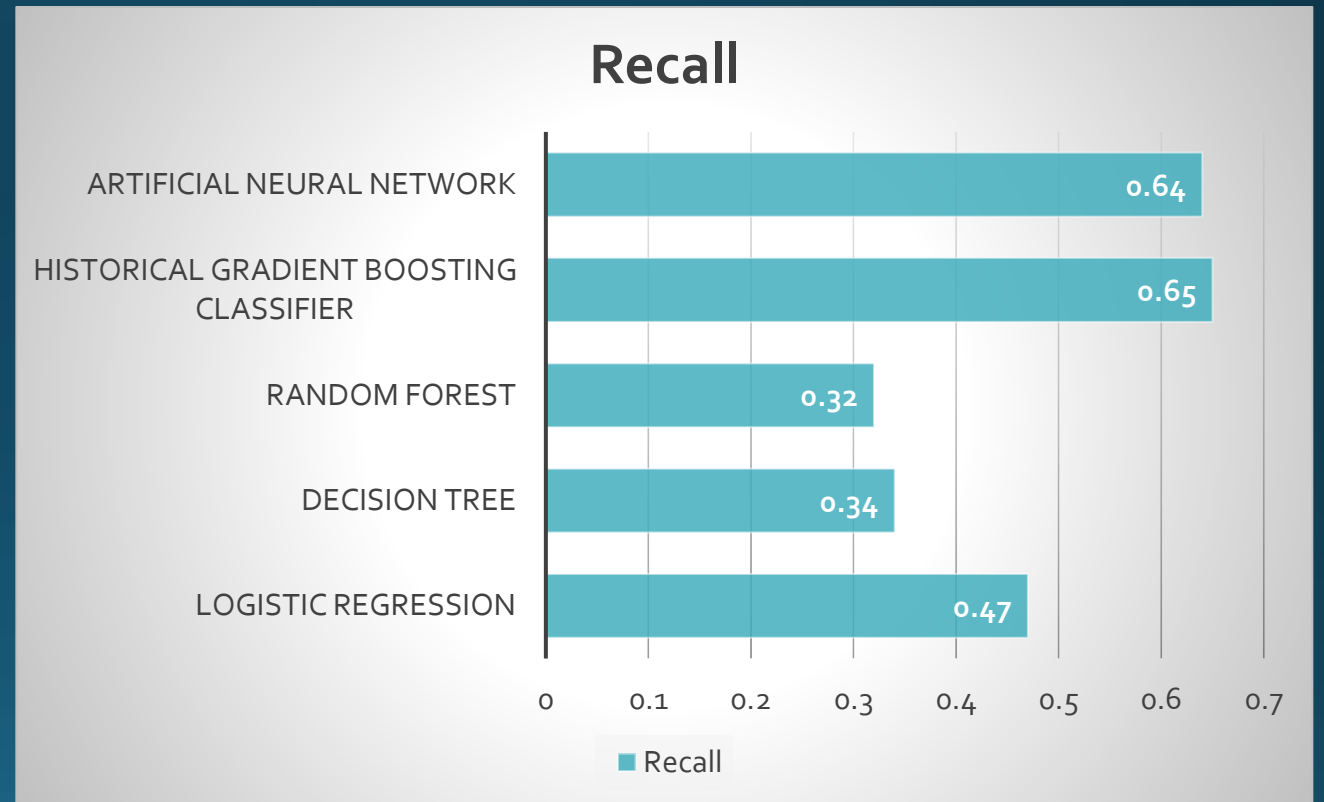
Scaling

Critical Recall: Model Testing

Simple models like Logistic Regression (47%) actually beat the basic tree models (~33%)

But the boosted and neural models cracked the complexity.

0.02 gap between training and testing accuracy





2 of 3 long-term residents can be
identified on day 1